Sheth L.U.J. And Sir M.V. College

Practical No. 15

15 Generating basic summaries using str() or summary() (R).

Output:-

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> tobacco_df <- read.csv("tobacco.csv", na.strings = c("", "NA"))
> print("--- Data Loaded ---")
[1] "--- Data Loaded ---"
> print(head(tobacco_df))
  id kode_provinsi nama_provinsi kode_kabupaten_kota   nama_kabupaten_kota jumlah_produksi satuan tahun
1  1            32    JAWA BARAT                3201      KABUPATEN BOGOR               0    TON  2016
2  2            32    JAWA BARAT                3202    KABUPATEN SUKABUMI               0    TON  2016
3  3            32    JAWA BARAT                3203     KABUPATEN CIANJUR              46    TON  2016
4  4            32    JAWA BARAT                3204     KABUPATEN BANDUNG            1358    TON  2016
5  5            32    JAWA BARAT                3205       KABUPATEN GARUT            3597    TON  2016
6  6            32    JAWA BARAT                3206 KABUPATEN TASIKMALAYA              30    TON  2016
> print("--- OUTPUT OF str() ---")
[1] "--- OUTPUT OF str() ---"
> str(tobacco_df)
'data.frame':   243 obs. of  8 variables:
 $ id                 : int  1 2 3 4 5 6 7 8 9 10 ...
 $ kode_provinsi      : int  32 32 32 32 32 32 32 32 32 32 ...
 $ nama_provinsi      : chr  "JAWA BARAT" "JAWA BARAT" "JAWA BARAT" "JAWA BARAT" ...
 $ kode_kabupaten_kota: int  3201 3202 3203 3204 3205 3206 3207 3208 3209 3210 ...
 $ nama_kabupaten_kota: chr  "KABUPATEN BOGOR" "KABUPATEN SUKABUMI" "KABUPATEN CIANJUR" "KABUPATEN BANDUNG" ...
 $ jumlah_produksi    : int  0 0 46 1358 3597 30 46 342 0 737 ...
 $ satuan             : chr  "TON" "TON" "TON" "TON" ...
 $ tahun              : int  2016 2016 2016 2016 2016 2016 2016 2016 2016 2016 ...
> print("--- OUTPUT OF summary() [Before Factor Conversion] ---")
[1] "--- OUTPUT OF summary() [Before Factor Conversion] ---"
> summary(tobacco_df)
       id          kode_provinsi nama_provinsi       kode_kabupaten_kota nama_kabupaten_kota jumlah_produksi      satuan
```

```
Console   Terminal ×   Background Jobs ×
R 4.5.2 · ~/
> summary(tobacco_df)
       id          kode_provinsi nama_provinsi      kode_kabupaten_kota nama_kabupaten_kota jumlah_produksi     satuan
 Min.   :  1.0   Min.   :32    Length:243         Min.   :3201        Length:243          Min.   :   0.0   Length:243
 1st Qu.: 61.5   1st Qu.:32    Class :character   1st Qu.:3207        Class :character    1st Qu.:   0.0   Class :character
 Median :122.0   Median :32    Mode  :character   Median :3214        Mode  :character    Median :   0.0   Mode  :character
 Mean   :122.0   Mean   :32                       Mean   :3231                            Mean   : 312.1
 3rd Qu.:182.5   3rd Qu.:32                       3rd Qu.:3273                            3rd Qu.:  42.5
 Max.   :243.0   Max.   :32                       Max.   :3279                            Max.   :3597.0
     tahun
 Min.   :2016
 1st Qu.:2018
 Median :2020
 Mean   :2020
 3rd Qu.:2022
 Max.   :2024
> factor_cols <- c("nama_provinsi", "nama_kabupaten_kota", "satuan")
> for(col in factor_cols){
+   if(col %in% names(tobacco_df)){
+     tobacco_df[[col]] <- as.factor(tobacco_df[[col]])
+   }
+ }
> print("--- OUTPUT OF summary() [After Factor Conversion] ---")
[1] "--- OUTPUT OF summary() [After Factor Conversion] ---"
> summary(tobacco_df)
       id          kode_provinsi     nama_provinsi kode_kabupaten_kota               nama_kabupaten_kota jumlah_produksi    satuan
 Min.   :  1.0   Min.   :32      JAWA BARAT:243  Min.   :3201        KABUPATEN BANDUNG        :  9      Min.   :   0.0   TON:243
 1st Qu.: 61.5   1st Qu.:32                      1st Qu.:3207        KABUPATEN BANDUNG BARAT:  9       1st Qu.:   0.0
```


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Subject:- Data Analysis With SAS/SPSS/R

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Practical No. 15

```
> summary(tobacco_df)
       id         kode_provinsi    nama_provinsi  kode_kabupaten_kota                nama_kabupaten_kota jumlah_produksi     satuan
 Min.   :  1.0   Min.   :32     JAWA BARAT:243   Min.   :3201         KABUPATEN BANDUNG      :  9      Min.   :   0.0    TON:243
 1st Qu.: 61.5   1st Qu.:32                      1st Qu.:3207         KABUPATEN BANDUNG BARAT:  9      1st Qu.:   0.0
 Median :122.0   Median :32                      Median :3214         KABUPATEN BEKASI       :  9      Median :   0.0
 Mean   :122.0   Mean   :32                      Mean   :3231         KABUPATEN BOGOR        :  9      Mean   : 312.1
 3rd Qu.:182.5   3rd Qu.:32                      3rd Qu.:3273         KABUPATEN CIAMIS       :  9      3rd Qu.:  42.5
 Max.   :243.0   Max.   :32                      Max.   :3279         KABUPATEN CIANJUR      :  9      Max.   :3597.0
                                                                      (Other)                :189
     tahun
 Min.   :2016
 1st Qu.:2018
 Median :2020
 Mean   :2020
 3rd Qu.:2022
 Max.   :2024

> avg_production <- mean(tobacco_df$jumlah_produksi, na.rm = TRUE)
> max_production <- max(tobacco_df$jumlah_produksi, na.rm = TRUE)
> total_production <- sum(tobacco_df$jumlah_produksi, na.rm = TRUE)
> min_year <- min(tobacco_df$tahun, na.rm = TRUE)
> max_year <- max(tobacco_df$tahun, na.rm = TRUE)
> most_common_province <- names(
+   sort(table(tobacco_df$nama_provinsi), decreasing = TRUE)
+ )[1]
> most_common_kabupaten <- names(
+   sort(table(tobacco_df$nama_kabupaten_kota), decreasing = TRUE)
+ )[1]
```

```
> avg_production <- mean(tobacco_df$jumlah_produksi, na.rm = TRUE)
> max_production <- max(tobacco_df$jumlah_produksi, na.rm = TRUE)
> total_production <- sum(tobacco_df$jumlah_produksi, na.rm = TRUE)
> min_year <- min(tobacco_df$tahun, na.rm = TRUE)
> max_year <- max(tobacco_df$tahun, na.rm = TRUE)
> most_common_province <- names(
+   sort(table(tobacco_df$nama_provinsi), decreasing = TRUE)
+ )[1]
> most_common_kabupaten <- names(
+   sort(table(tobacco_df$nama_kabupaten_kota), decreasing = TRUE)
+ )[1]
> print(paste("Average Tobacco Production:", avg_production))
[1] "Average Tobacco Production: 312.053497942387"
> print(paste("Maximum Tobacco Production:", max_production))
[1] "Maximum Tobacco Production: 3597"
> print(paste("Total Tobacco Production:", total_production))
[1] "Total Tobacco Production: 75829"
> print(paste("Data Available From Year:", min_year, "to", max_year))
[1] "Data Available From Year: 2016 to 2024"
> print(paste("Most Frequent Province:", most_common_province))
[1] "Most Frequent Province: JAWA BARAT"
> print(paste("Most Frequent Kabupaten/Kota:", most_common_kabupaten))
[1] "Most Frequent Kabupaten/Kota: KABUPATEN BANDUNG"
>
>
```


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